While there has been progress on specific details required by this vision (such as trying to understand why only three of the universe's spatial dimensions underwent expansion, as well as developing mathematical methods that may prove relevant to analyzing the spaceless/timeless realm that may precede inflation), the eureka moments have yet to occur. The intuition is that whereas inflationary cosmology imagines the observable universe getting ever smaller at ever earlier times—and hence being ever hotter, denser, and energetic—string/M-theory tames this unruly (in physics-speak, "singular") behavior by introducing a minimal size (as in our discussion on pages 350–351) below which new and less singular physical quantities become relevant. This reasoning lies at the heart of string/M-theory's successful merger of general relativity and quantum mechanics, and many researchers expect that we will shortly determine how to apply the same reasoning in the context of cosmology. But, as of now, the fuzzy patch still looks fuzzy, and it's anybody's guess when clarity will be achieved.

The second approach employs the braneworld scenario, and in its most radical incarnation posits a completely new cosmological framework. It is far from clear whether this approach will survive detailed mathematical scrutiny, but it does provide a good example of how breakthroughs in fundamental theory can suggest novel trails through well-trodden territory. The proposal is called the *cyclic model.*

Cyclic Cosmology

From the standpoint of time, ordinary experience confronts us with two types of phenomena: those that have a clearly delineated beginning, middle, and end (this book, a baseball game, a human life) and those that are cyclic, happening over and over again (the changing seasons, the rising and setting of the sun, Larry King's weddings). Of course, on closer scrutiny we learn that cyclic phenomena also have a beginning and end, since cycles do not generally go on forever. The sun has been rising and setting—that is, the earth has been spinning on its axis while revolving around the sun—every day for some 5 billion years. But before that, the sun and the solar system had yet to form. And one day, some 5 billion years from now, the sun will turn into a red giant star, engulfing